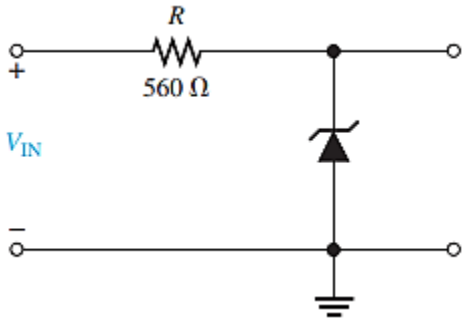
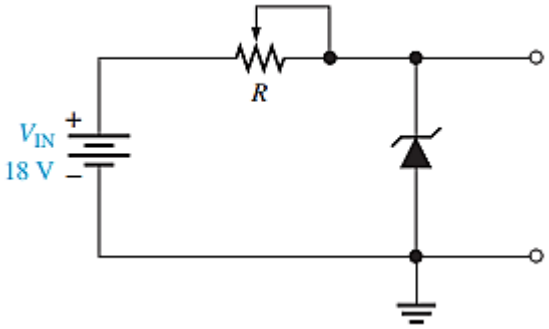
## Sheet 6

1. A Zener has an impedance of 15 Ω. What is its terminal voltage at 50 mA if $V_Z$ = 4.7 V at $I_Z$ = 25 mA?
2. Determine the minimum input voltage required for regulation to be established in the figure below. Assume an ideal Zener diode with $I_{ZK}$ = 1.5 mA and $V_Z$ = 14 V.



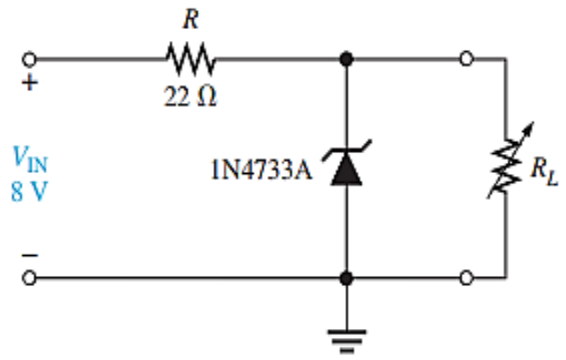

3. Repeat Problem (2) with $Z_Z$ = 20 Ω and $V_Z$ = 14 V at 30 mA

4. To what value must *R* be adjusted in the following figure to make $I_Z$ = 40 mA? Assume $V_Z$ = 12 V at 30 mA and $Z_Z$ = 30 Ω




5. If A 20 V peak sinusoidal voltage is applied to the circuit in problem (4) in place of the dc source. Draw the output waveform. Use the parameter values established in Problem 4.

***6.*** A loaded Zener regulator is shown in the following figure. Determine the minimum and maximum permissible load currents. $Z_Z$ = 7 Ω, and $I_{ZM}$ = 70 mA. $V_Z$ = 5.1 V at $I_Z$ = 49 mA, $I_{ZK}$ = 1 mA.




**7.** Find the load regulation expressed as a percentage in Problem 6

**8.** Find the minimum load resistance (Ω) for the shown network.

